

Reading 3: Due Wednesday

Assignment 3: Due October 18

Test 1: Wednesday after reading week on Chapters 1 & 2

§ 2.2 Set Operations

$\cap$ - intersection

$\cup$ - union

$\setminus$ - set difference

Definiton

if S and T are sets i

1. $S \cap T$ is the intersection of S with T consisting of elements that are in both S and T

• i.e $x \in S \cap T \leftrightarrow x \in S \wedge x \in T$

2. $S \cup T$ is the union of S or T

• i.e $x \in S \cup T \leftrightarrow x \in S \vee x \in T$

3. $S \setminus T$ (sometimes is $S - T$) is the set difference $S \setminus T = \{x \mid x \in S \wedge x \notin T\}$

4. With respect to some universal set u , $\bar{S}$ is $U \setminus S$

Example

If $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$ then

$$A \cap B = \{2, 4\}$$

$$A \cup B = \{1, 2, 3, 4, 6\}$$

$$A \setminus B = \{1, 3\} \text{ but } B \setminus A = \{6\}$$

Observe, there is no obvious relation between $A \setminus B$ and $B \setminus A$

Note:

$$|A \cap B| + |A \setminus B| + |B \setminus A| = 2 + 2 + 1 = 5 = |A \cup B|$$

This works for any finite sets,

$$\text{If } U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$\text{then } \bar{A} = \{5, 6, 7, 8\} \text{ and } \bar{B} = \{1, 3, 5, 7, 8\}$$

$$1. \bar{A} \cap \bar{B} = \{5, 7, 8\}$$

and also

$$2. \overline{A \cup B} = \{5, 7, 8\}$$

1. and 2. are specific sets depend on U , try with $U = \mathbb{Z}$ or $U = \{1, 2, 3, 4, 5, 6, \}$

The relations

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

and

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

apply for all sets and all universes (also De Morgan's Law)

How to see this? (questions)

Turn it into a logic question (answer)

If

$$x \in \overline{A \cap B} \text{ then } x \notin A \cap B$$

So

$$\neg(x \in A \cap B)$$

or equivalently

$$\neg(x \in A \wedge x \in B)$$

$$\equiv x \notin A \vee x \notin B$$

$$\equiv x \in \overline{A} \vee x \in \overline{B}$$

$$\equiv x \in \overline{A} \cup \overline{B}$$

The above shows $x \in \overline{A \cap B} \rightarrow x \in \overline{A} \cup \overline{B}$

Thus

$$\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$$

We also need to show:

$$x \in \overline{A} \cup \overline{B} \rightarrow x \in \overline{A \cap B}$$

Set identities can be verified using set membership tables which are like truth tables, but use 1 or 0 instead of T and F and have as rows the regions of an appropriate Venn Diagram

Venn Diagrams

With more than 3 sets, these get messy.

We can also build new identities from old ones

Example

$$(A \cap B) \cup C = C \cup (A \cap B) \text{ (by commutative Law)}$$

$$= (C \cup A) \cap (C \cup B) \text{ (by distributive Law)}$$

$$= (A \cup C) \cap (B \cup C) \text{ (by commutative Laws)} = (A \cup C) \cap (B \cup C) \text{ (by commutative Laws)}$$

§ 2.3 Functions

More like computer science functions than calculus functions. (Not drawing graphs, or slopes, etc)

Definition

A function f from set A to set B is a rule for assigning an element of B to each element of A , we write:

$f : A \rightarrow B$ (f is a function from A to B)

f is the name of the function

A is the domain of the function f

B is the codomain of the function f (the range is the set of actual outputs.)

Example

If A is all the students in this room

B is a set of colors

and

f is the favorite color function, then the range of f is

{lavender, turquoise, blue, red, green, white, teal, purple, ...}

Note: Puse is not in the range, but is in the codomain

Notation

If the value $y(\in B)$ is assigned to $x(\in A)$ then we write:

$$f(x) = y$$

y is called the image of x

x is called a pre-image of y

Sometimes we apply a function to a set,

$$f(S) = \{x \in B \mid \exists a \in S(f(a) = x)\}$$

Example

If $S = \{\text{Net, Ali, Rayan}\}$

the $f(S) = \{\text{teal, blue}\}$

The range of f is $f(A)$ where A is the domain,

Nice Functions

We will only talk about ‘nice’ functions, since most functions can’t be described at all.

Some ways functions can be nice is if they interact ‘nicely’ with their codomains.

Types of Functions

- onto functions
- one-to-one functions
- one-to-one correspondences

Alternate names (from their types)

- Surjections
- Injections
- Bijections